

RF Design Course 2026

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1 Task 14

1.1 Number 5

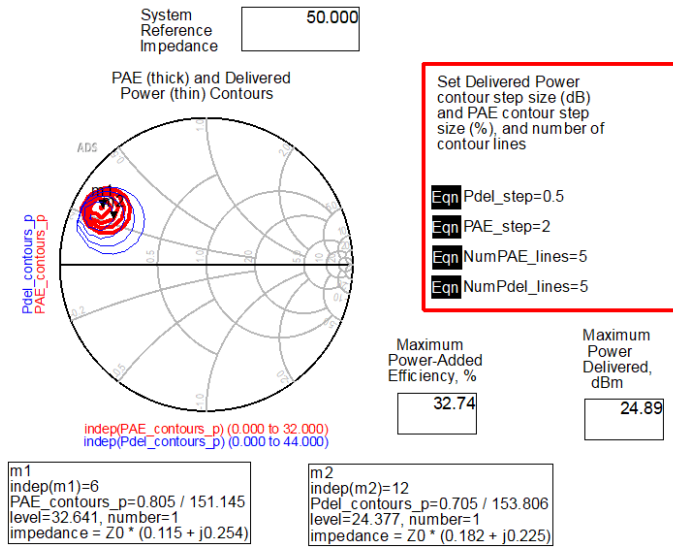


Figure 1: P_{del} Plot for $V_{high} = 6.5V$ and $V_{low} = 2V$

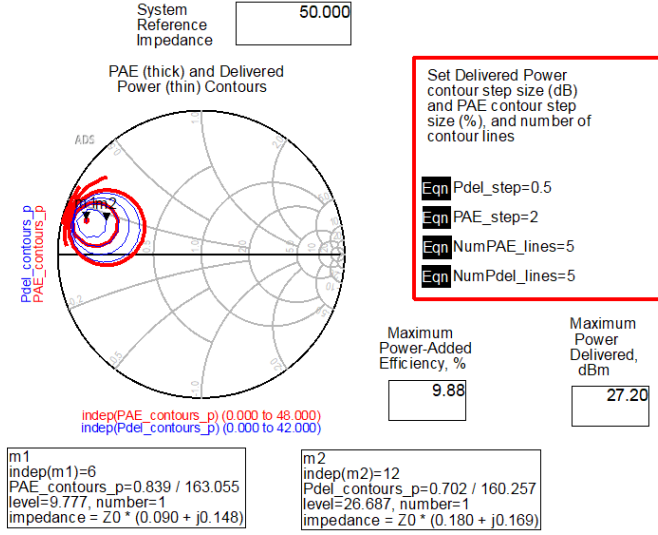


Figure 2: P_{del} Plot for $V_{high} = 6.5V$ and $V_{low} = 3V$

1.2 Number 6

- Increasing V_{low} from 2V to 3V raises the gate bias from near Class-B to Class-A/B operation, which dramatically reduces the maximum Power Added Efficiency from 32.74% to 9.88% because the higher quiescent current increases DC power consumption significantly.
- The maximum delivered power increases from 24.89 dBm to 27.20 dBm when V_{low} is increased to 3V, demonstrating that Class-A/B biasing allows for higher output power capability at the expense of efficiency.
- The optimum load impedance for maximum PAE changes from $Z_0(0.115 + j0.254)$ at $V_{low}=2V$ to $Z_0(0.090 + j0.148)$ at $V_{low}=3V$, indicating that Class-B operation requires a more inductive load to achieve its peak efficiency.
- The optimum impedance for maximum delivered power remains relatively consistent between both bias points ($Z_0(0.182 + j0.225)$ vs $Z_0(0.180 + j0.169)$), confirming that the real part of the optimum load is determined more by the drain voltage than by the gate bias.
- At $V_{low}=2V$ (Class-B), the amplifier achieves its maximum PAE of 32.74% at a point where the delivered power is only slightly below its maximum, whereas at $V_{low}=3V$ (Class-A/B), the 9.88% PAE represents a significant efficiency penalty with only a modest power gain of 2.31 dB.

1.3 Number 7

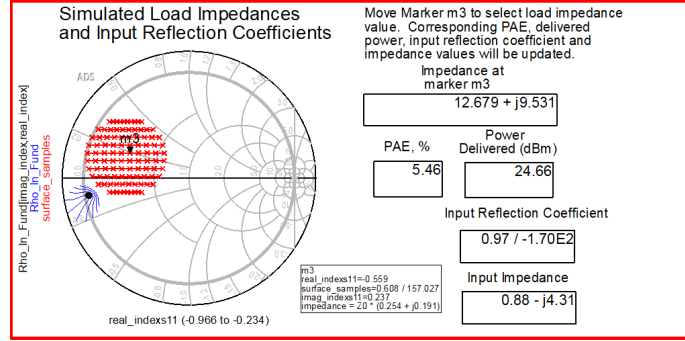


Figure 3: For frequency = 850 MHz, Pavs = 7 dB and MOS = 28

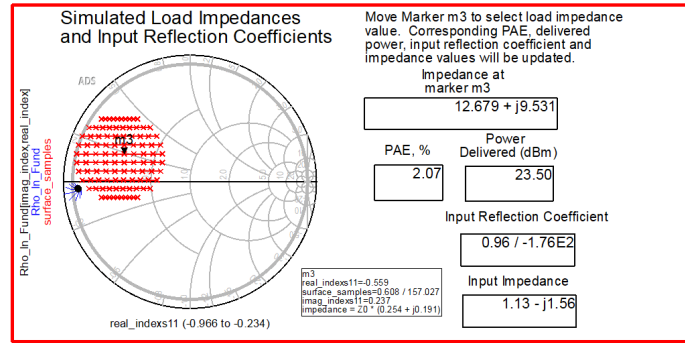


Figure 4: For frequency = 850 MHz, Pavs = 7 dB and MOS = 56

- Sentences on Behavior:
Doubling the number of MOS cells from 28 to 56 approximately halves the input impedance magnitude because the input capacitance (C_{gs}) doubles, following the same principle as parallel capacitors. The input power is increased by 3 dB (from 7 dBm to 10 dBm) to maintain the same drive per cell since the device now has twice the power handling capability. Despite the larger device size, the delivered power remains similar because the load impedance is kept constant and has not been re-optimized for the new cell count.
- Sentences on Parallel Capacitors:
When two identical capacitors are connected in parallel, the total capacitance doubles ($C_{total} = 2C$) and the impedance halves ($Z_{total} = Z/2$) because they share the same voltage but split the current equally. Similarly, the gate-source capacitance of each MOS cell adds in parallel, so doubling

the number of cells doubles the total input capacitance and halves the input impedance magnitude.

1.4 Number 8

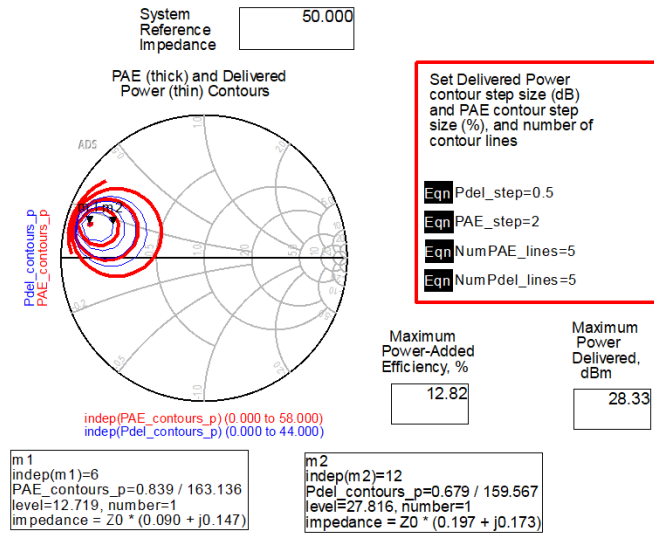


Figure 5: Load Impedance for Pavs = 10dB, Cells = 28, Vhigh = 6.5V and Vlow = 3V

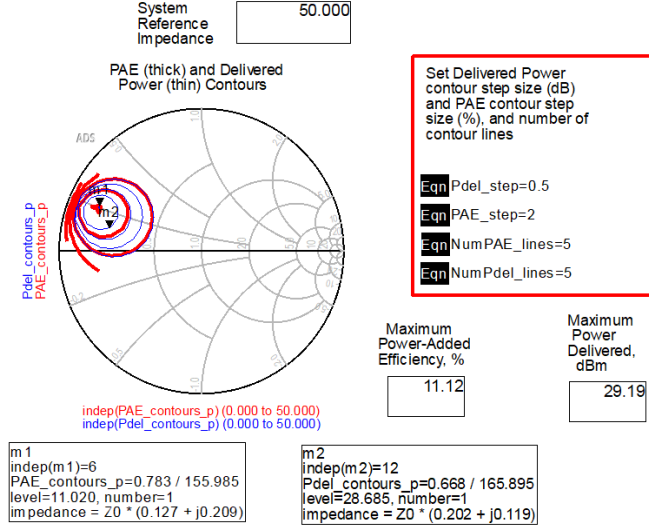


Figure 6: Load Impedance for Pavs = 10dB, Cells = 28, Vhigh = 9V and Vlow = 3V

- Increasing Vhigh from 6.5V to 9V increases the maximum delivered power by 0.86 dB (from 28.33 dBm to 29.19 dBm) because the higher drain voltage allows for a larger voltage swing at the output and a longer load line before the device enters saturation.
- The maximum Power Added Efficiency decreases from 12.82% to 11.12% when Vhigh is increased to 9V because the higher drain bias significantly increases DC power consumption ($P_{dc} = V_{ds} \times I_{ds}$), reducing the efficiency despite the modest improvement in output power.
- The real part of the optimum load impedance for maximum PAE increases from 4.50 Ω to 6.35 Ω when Vhigh is raised to 9V, which is consistent with the behavior of a MOSFET in saturation where the output resistance (r_{ds}) increases with higher drain-to-source voltage (V_{ds}).
- The real part of the optimum load impedance for maximum delivered power shows only a slight increase from 9.85 Ω to 10.10 Ω , confirming that the power-optimized load is primarily determined by the load-line resistance calculated from $(V_{ds} - V_{knee})/I_{ds}$ and is relatively stable across moderate V_{ds} changes.
- The optimum impedance for PAE becomes significantly more inductive at 9V (from $+j7.35 \Omega$ to $+j10.45 \Omega$) while the optimum for power becomes less inductive (from $+j8.65 \Omega$ to $+j5.95 \Omega$), indicating that the trade-off between power and efficiency requires different reactive terminations as the bias voltage changes.

1.5 Number 9

- **Efficiency Equations:** The formal definitions for Drain Efficiency (DE) and Power-Added Efficiency (PAE) are given by:

$$\text{Drain Efficiency (DE)} = \frac{P_{out}}{P_{DC}} \times 100\%$$

$$\text{Power-Added Efficiency (PAE)} = \frac{P_{out} - P_{in}}{P_{DC}} \times 100\%$$

- **Drain Efficiency (DE) Explanation:** Drain Efficiency (DE) measures how efficiently the transistor converts DC power from the power supply into RF output power. It is calculated as the ratio of RF output power to DC power consumption (P_{DC}), expressed as a percentage. Crucially, DE only considers the output side and does not account for the input drive power required to excite the device.
- **Power-Added Efficiency (PAE) Explanation:** Power-Added Efficiency (PAE) provides a more complete measure of an amplifier's true physical efficiency because it explicitly accounts for both the input and output power levels. By subtracting P_{in} from P_{out} , it isolates the actual "added" power generated by the transistor before normalizing it against the DC power consumption. Because P_{in} is subtracted from the numerator, PAE will always be mathematically lower than DE.
- **Mathematical Relationship:** For a power amplifier exhibiting an operating power gain of $G = \frac{P_{out}}{P_{in}}$, the direct relationship between the two metrics can be written as:

$$\text{PAE} = \text{DE} \times \left(1 - \frac{1}{G}\right)$$

- **Design Implications:** As the gain (G) of the stage increases, the fractional term $\left(1 - \frac{1}{G}\right)$ asymptotically approaches 1, causing the PAE value to converge toward the DE value. For high-gain high-frequency amplifiers, PAE and DE tracking remain very close in total magnitude, which explains why both performance metrics yield similar final percentages in this specific design sweep example.

1.6 Number 12

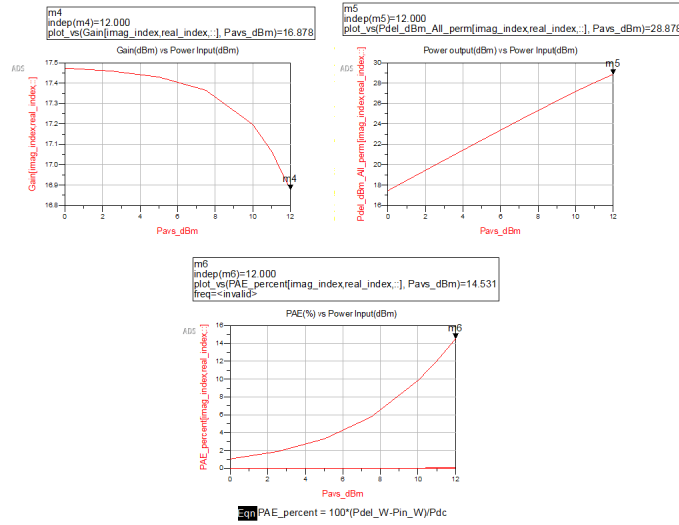


Figure 7: For RFfreq = 1000 MHz, Vhigh = 6.5V and Vlow = 3V

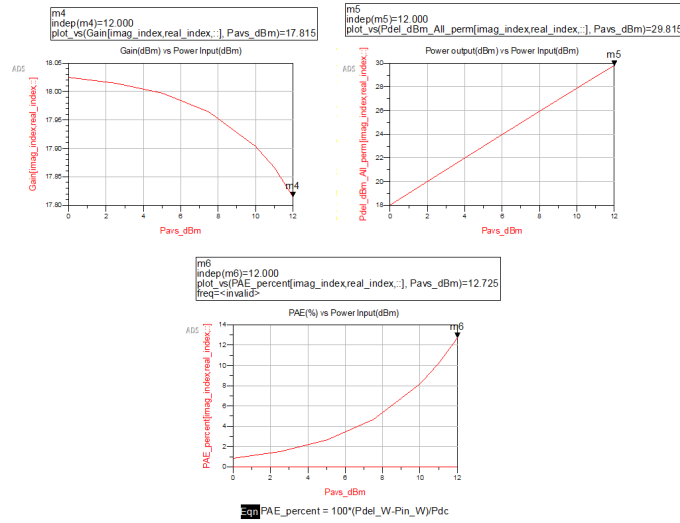


Figure 8: For RFfreq = 1000 MHz, Vhigh = 9V and Vlow = 3V

1.7 Number 13

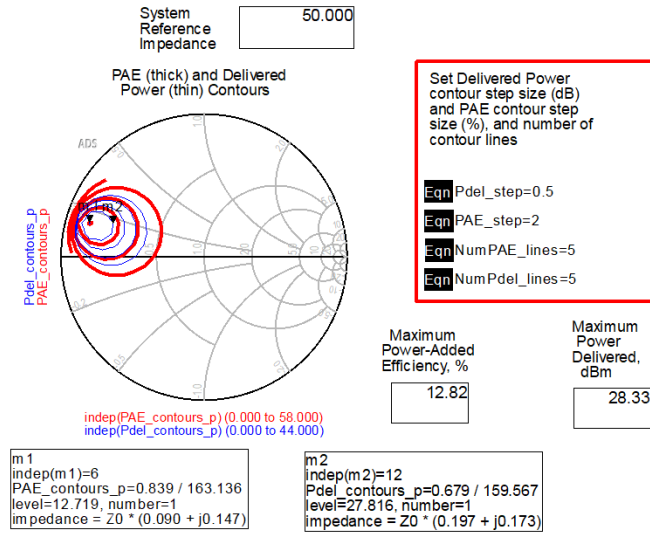


Figure 9: Load Impedance for Pavs = 10dB, Cells = 28, Vhigh = 6.5V and Vlow = 3V

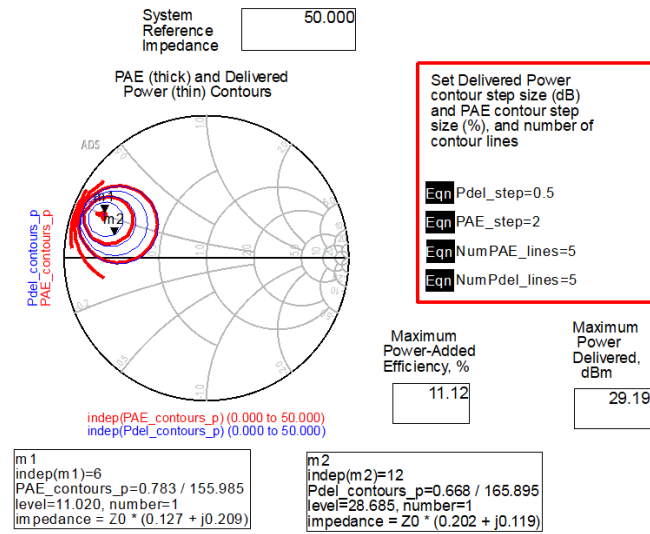


Figure 10: Load Impedance for Pavs = 10dB, Cells = 28, Vhigh = 9V and Vlow = 3V

- No, we do not use the same load impedance for both V_{high} values. The optimum load impedance for maximum PAE changes from $Z = 50 \times (0.090 + j0.147) = 4.50 + j7.35 \Omega$ at $V_{high} = 6.5$ V to $Z = 50 \times (0.127 + j0.209) = 6.35 + j10.45 \Omega$ at $V_{high} = 9$ V. The real part of the optimum load impedance increases with higher V_{high} because the output resistance (r_{ds}) of the MOSFET in the saturation region increases with drain-to-source voltage, requiring a higher load resistance for optimal power transfer and efficiency. This is consistent with large-signal load-line theory where:

$$R_{opt} = \frac{V_{ds} - V_{knee}}{I_{d,max}/2}$$

Therefore, a higher V_{ds} directly leads to a higher optimum load resistance. The higher V_{ds} (9 V) allows for a longer load line and a larger peak-to-peak voltage swing, which shifts the optimum load impedance to a higher real part, while the reactive part (inductive component) also increases to compensate for the shifted device output capacitances at this elevated bias voltage. Ultimately, this demonstrates the fundamental trade-off in RF power amplifier design: a higher drain voltage yields superior peak output power capability but strictly requires a different fundamental load impedance and results in slightly lower overall efficiency metrics.

2 Task 16

Settings

- $V_{high} = 6$ V, $V_{low} = -3$ V
- $\text{Mag}_\rho = 0.600$, $\Phi_\rho = 150^\circ$
- $Z_L \approx 13.338 + j12.504 \Omega$
- $Z_{L,2} = 1000 + j0 \Omega$ (open), $Z_{L,3} = 1000 + j0 \Omega$ (open)

Plots

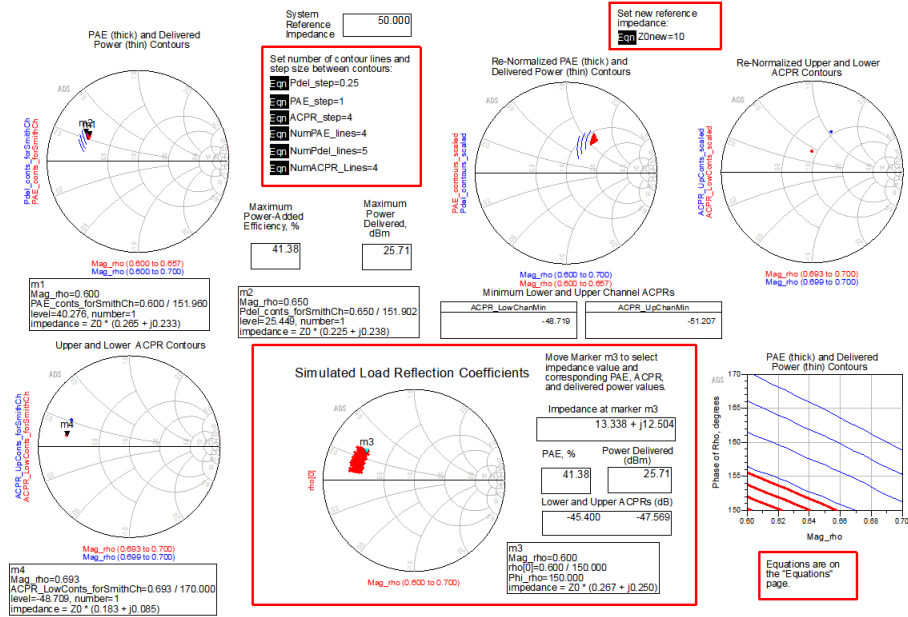


Figure 11: Plot for 700 MHz and $P_{avs} = 16.0 \text{ dBm}$

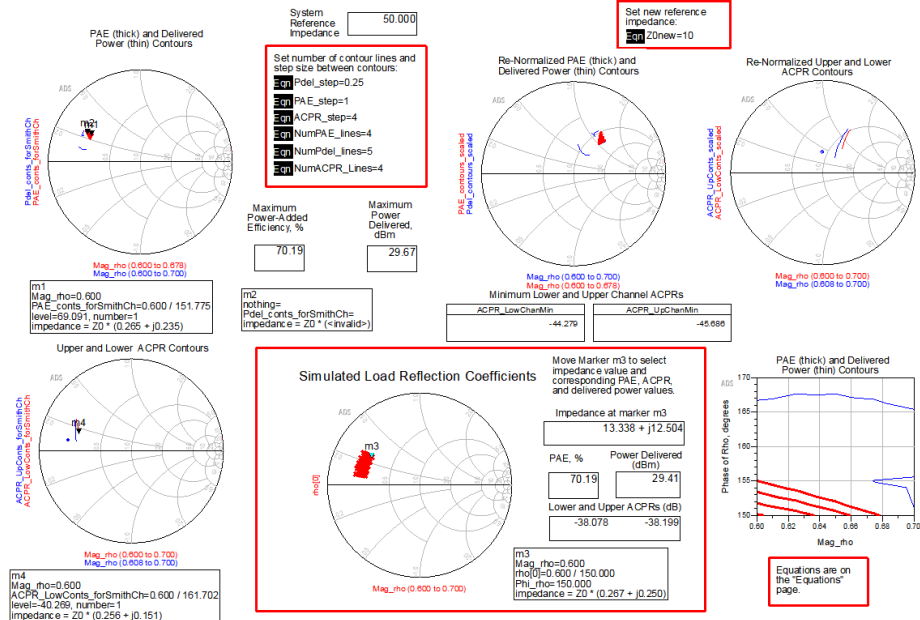


Figure 12: Plot for 700 MHz and $P_{avs} = 22.75\text{dBm}$

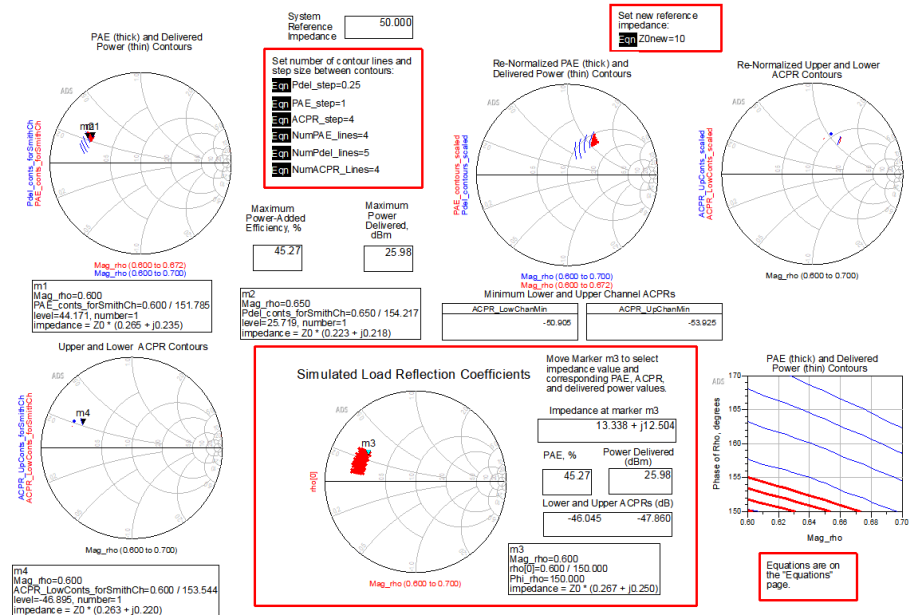


Figure 13: Plot for 900 MHz and $P_{avs} = 17.5\text{dBm}$

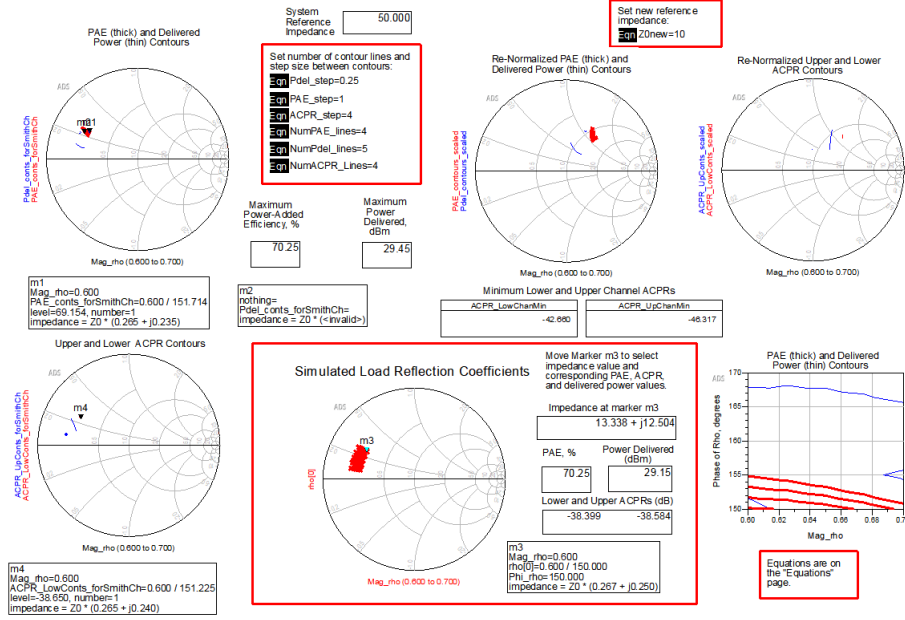


Figure 14: Plot for 900 MHz and $P_{avs} = 23.5\text{dBm}$

Table 1: Performance Summary at Different Frequencies and Target Case Conditions

Case	Frequency	P_{avs}	PAE	P_{out}	ACPR low	ACPR up
-45 dBc	700 MHz	16.0 dBm	41.38%	25.71 dBm	-45.400	-47.569
-45 dBc	900 MHz	17.5 dBm	45.27%	25.98 dBm	-46.045	-47.860
-38 dBc	700 MHz	22.75 dBm	70.19%	29.41 dBm	-38.078	-38.199
-38 dBc	900 MHz	23.5 dBm	70.25%	29.15 dBm	-38.399	-38.584

Figure of Merit (FOM)

-45 dBc

$$FOM_{45} = \frac{41.38 \cdot 700 + 45.27 \cdot 900}{2}$$

$$FOM_{45} = \frac{28966 + 40743}{2} = 34854.5$$

-38 dBc

$$FOM_{38} = \frac{70.19 \cdot 700 + 70.25 \cdot 900}{2}$$

$$FOM_{38} = \frac{49133 + 63225}{2} = 56179$$

Overall FOM

$$FOM = \frac{34854.5 + 56179}{2} = 45516.75$$

$$\boxed{FOM \approx 45517}$$